Calculus mod 5

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About

These are some lecture notes for a series of talks given at the Chicago REU in the summer of 2026. I'm very grateful to Peter May, who organized the REU and let me speak. I also want to thank Kenta Suzuki; he and I run a blog 'hidden-phenomena', and lecture 2 of these notes is an expansion of an article from this blog.

Contents

Lecture 1

What is algebraic geometry?

1.1 Introduction

Algebraic geometry is the study of shapes defined by systems of polynomial equations.

What makes it interesting to single out the class of shapes defined by polynomial equations in particular? In other words – why is algebraic geometry its own subject, and not just a source of examples for algebraic topology and Riemannian geometry courses?

There are several answers one could give to this question. The point of view taken by this course is that these shapes are interesting because they make sense over many different number systems.

To illustrate what we mean, let's single out the parabola $y = x^2$. There are many different ways one could interpret it:

1. if x, y are interpreted as *real numbers*, we get the standard parabola shape in $\mathbb{R}^2$;

2. if x, y are *complex numbers*, then $y = x^2$ defines a certain 2-dimensional subset of the 4-dimensional space $\mathbb{C}^2$;

3. if x, y are interpreted as *integers modulo 3*, then the parabola is the 'shape' consisting of the three points $(0, 0), (1, 1)$, and $(2, 1)$;

4. if x, y are interepreted as *integers*, then the parabola is essentially the set of all perfect squares: it contains points like $(5, 25), (13, 169)$, and $(-3, 9)$, where the x-coordinate can be any integer, and y is always its square.

The miracle of algebraic geometry is that these different incarnations of the shape $y = x^2$ are somehow related. The purpose of algebraic geometry, from this point of view, is to develop a language with which one can talk about all of these different incarnations at once, and therefore reason about the relationships between them.

Example 1.1.1 (Integrals and Diophantine equations)**.** A high school calculus student can integrate

$$\int \frac{dx}{\sqrt{1-x^2}}. \tag{1.1.2}$$

However, the similar looking integral

$$\int \frac{dx}{\sqrt{x^3-2}} \tag{1.1.3}$$

would give even an expert undergraduate calculus student trouble!

When approaching the first integral, it is tempting to substitute $x = \sqrt{1-y^2}$. Then x and y obey the algebraic relationship

$$x^2 + y^2 = 1.$$

This allows you to evaluate (1.1.2) in terms of trigonometric functions (in fact, (1.1.2) evaluates to $\arcsin(y) + C$).

Similarly, the second integral is related to the algebraic curve

$$y^2 = x^3 - 2.$$

These two curves, $x^2+y^2 = 1$ and $y^2 = x^3-2$, are related to the Diophantine equations

$$a^2 + b^2 = c^2 \tag{1.1.4}$$

and

$$a^2c = b^3 - 2c^3. \tag{1.1.5}$$

It's very easy to find (with proof!) all the integer solutions to (1.1.4); this was done by the ancient Greek mathematician Euclid; and even before Euclid, the mathematician Diophantus found an infinite family of solutions, and the Babylonians found many particular solutions.

However, (1.1.5) is a much trickier Diophantine equation. It has essentially one solution (observe $25 = 27 - 2$), but proving this is quite tricky unless you know some algebraic number theory!

Perhaps remarkably, the geometry of $x^2 + y^2 = 1$ explains why (1.1.2) is easy to solve, and why (1.1.4) is easy to solve; and the geometry of $y^2 = x^3 - 2$ explains why (1.1.3) and (1.1.5) are both hard to solve.

Example 1.1.6 (Dimension and point counts)**.** Consider the parabola $y = x^2$. Over the real numbers, the graph of $y = x^2$ gives a 1-dimensional object. And in modulo n arithmetic, there are exactly n pairs (x, y) which solve

$$y \equiv x^2 \pmod{n},$$

as x can be any residue modulo n, and then y is determined.

Consider now the polynomial equation

$$x + y + z = 0.$$

Over the real numbers, this defines a 2-dimensional plane in 3-dimensional space. And in modulo n arithmetic, there are exactly n^2 solutions to $x + y + z \equiv 0 \pmod{n}$.

Note that, in both cases, the real number incarnation of our shape had dimension d, and the 'modulo n' incarnation of our shape was a set of size n^d. The geometric notion of dimension is thus reflected arithmetically by the size of the solution set in modular arithmetic.

Remark 1.1.7. Dimension, intuitively, measures the 'number of free parameters' in a shape. If we are selecting d independent numbers in $\mathbb{Z}/n\mathbb{Z}$, then there are n^d total ways to make our selection, which explains why n^d corresponds to d-dimensions.

Example 1.1.8 (A more complicated dimension and point count example)**.** Consider the equation

$$x^2 + y^2 = 1. \tag{1.1.9}$$

Over $\mathbb{R}$, the solutions to (1.1.9) form a circle – a 1-dimensional shape.

How many solutions are there modulo n? Define

$$N_n := \text{number of solutions to } x^2 + y^2 \equiv 1 \pmod{n}.$$

Determining N_n is a fun exercise in elementary number theory[1]. When $n = p^k$ is the power of a prime number p, like $n = 289 = 17^2$ or $n = 243 = 3^5$, then N_n is easy to determine:

$$N_{p^k} = \begin{cases} p^k - p^{k-1} & p \equiv 1 \pmod{4}, \\ p^k + p^{k-1} & p \equiv 3 \pmod{4}, \\ 2p^k & p = 2, k > 1 \end{cases}.$$

From this, using the Chinese remainder theorem, we can deduce that $N_n = O(n)$; here, we're using big-O notation, meaning that N_n grows roughly as fast as n (and in particular, it grows slower than n^2 or n^3). This $O(n^1)$ is again related to the fact that the circle is 1-dimensional.

Example 1.1.10 (Point counting on the sphere)**.** In the vein of the previous example, consider the equation

$$x^2 + y^2 + z^2 = 1.$$

Over the real numbers, this defines the unit sphere, a 2-dimensional shape. One can use elementary number theory, similarly to the previous example, to show that the number of solutions to

$$x^2 + y^2 + z^2 \equiv 1 \pmod{n}$$

is $O(n^2)$. For example, when $n = p^k$ where $p \equiv 1 \pmod{4}$ is a prime number, then there are $p^{2k} - p^{2k-1} = \left(1 - \frac{1}{p}\right) n^2$ solutions. The 2 in this $O(n^2)$ reflects the 2-dimensional nature of the solution set.

[1] Hint: Use Legendre symbols!

1.2 Prespaces

To help formalize these ideas, we introduce now the notion of a *prespace*. A prespace is to algebraic geometry as a point-set topological space is to geometric topology: that is, all the objects of (classical) algebraic geometry are prespaces, but in practice algebraic geometers usually do not study arbitrary prespaces, and instead study prespaces obeying many extra properties. Compare this to how a geometric topologist does not study arbitrary point-set topological spaces, but usually assumes their spaces are Hausdorff, or metrizable, or locally Euclidean, etc.

In the introduction, we explained that the main benefit of studying shapes cut out by systems of polynomial equations is how they are realized over many different number systems. We are going to define a *prespace* as a shape which has an incarnation over every possible number system, where by number system we will mean *ring*.

Warning 1.2.1. In these notes, and in most notes on algebraic geometry, a ring is always assumed to be *commutative and unital*.

Definition 1.2.2. A *prespace* is a functor

$$X : \mathrm{Ring} \to \mathrm{Set}\,.$$

Remark 1.2.3 (What is a functor?)**.** For those who do not know category theory, when we say $X : \mathrm{Ring} \to \mathrm{Set}$ is a *functor*, we mean that,

1. for every ring A, we can produce a *set* $X(A)$, *and*

2. for every ring *homomorphism* $A \to B$, we can produce a *function* of sets $X(f) : X(A) \to X(B)$,

such that

1. whenever $f : A \to B$ and $g : B \to C$ are two ring homomorphisms, we have $X(g \circ f) = X(g) \circ X(f)$,

2. for any ring A, the function $X(\mathrm{id}_A)$ is the identity function on $X(A)$.

Remark 1.2.4. It is useful to think of mathematical definitions as consisting of two parts: data, and then properties which that data must obey. For example, a group consists of the data of a set and a binary operation, obeying certain properties (like associativity).

Similarly, the data of a functor is the sets $X(A)$ and the functions $X(f)$, and the property this data must obey is $X(g \circ f) = X(g) \circ X(f)$.

Definition 1.2.5. A *morphism* of prespaces

$$\eta : X \to Y$$

is a natural transformation of functors.

Remark 1.2.6 (What is a natural transformation?)**.** A morphism of prespaces is the data of, for every ring A, a function

$$\eta(A) : X(A) \to Y(A),$$

such that, for any ring homomorphism $\varphi : A \to B$, the diagram

$$\begin{array}{ccc} X(A) & \xrightarrow{\eta(A)} & Y(A) \\ \big\downarrow{\scriptstyle X(\varphi)} & & \big\downarrow{\scriptstyle Y(\varphi)} \\ X(B) & \xrightarrow{\eta(B)} & Y(B) \end{array}$$

commutes; that is, $Y(\varphi) \circ \eta(A) = \eta(B) \circ X(\varphi)$.

Remark 1.2.7. The reader should check that the following statements:

1. if $\eta : X \to Y$ and $\theta : Y \to Z$ are morphisms of prespaces, then the map $\theta \circ \eta : X \to Z$ defined by $(\theta \circ \eta)(A) := \theta(A) \circ \eta(A)$ is also a natural transformation;

2. a morphism of prespaces $\eta : X \to Y$ has an inverse $\eta^{-1} : Y \to X$ if, and only if, for every ring A, the function $\eta(A)$ is a bijection of sets.

Example 1.2.8 (The parabola)**.** We define the *parabola* as the prespace

$$P : \text{Ring} \to \text{Set}$$

given by

$$P(A) := \{(x, y) \in A \times A \mid y = x^2\}.$$

Given a ring homomorphism $\varphi : A \to B$, we define

$$P(\varphi) : P(A) \to P(B)$$

as the function

$$P(\varphi)(x, y) = (\varphi(x), \varphi(y)).$$

This pair $(\varphi(x), \varphi(y))$ belongs to $P(B)$ because, whenever $y = x^2$, we have

$$\varphi(y) = \varphi(x^2) = \varphi(x \cdot x) = \varphi(x) \cdot \varphi(x) = \varphi(x)^2,$$

as φ is a ring homomorphism.

Example 1.2.9 (Affine space)**.** We define *affine n-space* as the prespace

$$\mathbb{A}^n : \text{Ring} \to \text{Set}$$

given by

$$\mathbb{A}^n(A) := \{(x_1, ..., x_n) \in A^n\}.$$

1.3 Affine schemes

We now single out a very special class of prespaces.

Definition 1.3.1. Let R be a ring. The *affine scheme* associated to R is the prespace

$$\mathrm{Spec}(R) : \mathrm{Ring} \to \mathrm{Set}$$

given by

$$\mathrm{Spec}(R)(A) := \{\text{ring homomorphisms } \varphi : R \to A\}.$$

Example 1.3.2. Consider the ring $\mathbb{Z}[x_1, ..., x_n]$ of all polynomials in the variables $x_1, ..., x_n$. Then the affine scheme $\mathrm{Spec}(\mathbb{Z}[x_1, ..., x_n])$ coincides with the prespace $\mathbb{A}^n$ of Example 1.2.9; that is, there is an isomorphism of prespaces

$$\eta : \mathrm{Spec}(\mathbb{Z}[x_1, ..., x_n]) \simeq \mathbb{A}^n.$$

This isomorphism η is defined as follows. For a ring A, set

$$\eta(A) : \mathrm{Spec}(\mathbb{Z}[x_1, ..., x_n])(A) \to \mathbb{A}^n(A)$$

the function sending a ring homomorphism $\varphi : \mathbb{Z}[x_1, ..., x_n] \to A$ to the n-tuple $(\varphi(x_1), ..., \varphi(x_n)) \in \mathbb{A}^n(A)$.

We leave it to the reader to check that η defines a morphism of prespaces. To see it is an isomorphism (in light of Remark 1.2.7), it is enough to check that $\eta(A)$ is always a bijection; we leave this to the reader (it is the universal property of polynomial rings).

Remark 1.3.3. The parabola P of Example 1.2.8 is also an affine scheme:

$$P = \mathrm{Spec}(\mathbb{Z}[x, y]/(y - x^2)).$$

Can you see why?

Remark 1.3.4. Can you describe the set of ring homomorphisms

$$\mathbb{Z}[x, y, z]/(x^2 + y^2 + z^2 - 1) \to A,$$

for A an arbitrary ring, concretely?

Lecture 2

Tangent vectors and Hensel's lemma

2.1 Introduction

Calculus studies *approximations* to *continuous* quantities. Number theory typically involves studying *exact* answers to questions about *discrete* quantities. These seem, therefore, to be unrelated fields.

Yet Kurt Hensel made a remarkable discovery, showing that one could use ideas from calculus to solve number theory problems. We are going to explain Hensel's wonderful discovery by trying to solve polynomial equations in modular arithmetic. More particularly, we are going to try and find all integers x such that

$$x^3 - 17x^2 + 12x + 16 \equiv 0 \pmod{3000}. \tag{2.1.1}$$

While this sort of problem – solving a polynomial equation in modular arithmetic – might seem a bit strange, let me say a few things about why one would solve this.

2.1.1 Divisibility

One can profitably view the evolution of number theory through the lens of Diophantine equations. The first Diophantine equation you might try to solve is one of the form

$$Ax = B,$$

like $5x = 10$ or $2x = 7$. The theory of *integer divisibility* gives a completely satisfactory understanding of this class of Diophantine equations: you can solve $Ax = B$ exactly when B is divisible by A, in which case there is a unique integer solution, and there's a pretty efficient algorithm (long division!) which can be used to check divisibility.

To ask for the 'next simplest' Diophantine equation, there are two directions we could go. One could either increase the *degree* of the Diophantine equation, or increase the number of variables.

2.1.2 Increasing the degree

If we increase degrees, we're led to the *rational root theorem*, which gives a very simple algorithm for enumerating all integer solutions to any polynomial equation of the form

$$A_d x^d + A_{d-1} x^{d-1} + \cdots + A_0 = 0,$$

where the A_i are integers. If you're only interested in integer solutions, the rational root theorem is actually pretty easy, and only requires you to know about divisibility.

Proposition 2.1.2. *Any integer solution to*

$$A_d x^d + \cdots + A_0 = 0$$

must be a divisor of A_0.

Proof. If n is an integer solving this equation, then

$$A_0 = -n \cdot (A_1 + A_2 n + \cdots + A_d n^{d-1}),$$

and so $n \mid A_0$. $\square$

To enumerate all the divisors of A_0, you can do trial divisions, so it's not so bad (especially if you make the observation that you need to only check for divisors up to $\sqrt{A_0}$, as divisors come in pairs!).

2.1.3 Increasing the number of variables

The other way we could have made our Diophantine equation harder would be to increase the number of variables instead of the degree; in other words, we could try to solve an equation like

$$Ax + By = C. \tag{2.1.3}$$

This is an algebraic geometry course, not a number theory course, so let me just say that solving (2.1.3) is more or less equivalent to understanding the theory of *unique prime factorization* and *greatest common denominators.* Thus, an essential phenomenon in number theory – unique prime factorization – can be discovered by pondering a Diophantine equation.

2.1.4 To our question

To me, the point of mathematics is to *discover interesting mathematical phenomena*. I do not care about Diophantine equations or other math problems in their own right; I care about phenomena. However, it has been proven again and again throughout history that humans are really quite bad at observing mathematical phenomena in a background. We somehow need to think about an interesting problem in order for it to provoke us to discover something. So, every Diophantine equation, or other math problem, that I encounter, I view as a tool, with which I can use to either look for new phenomena (when I'm trying to solve an open problem for my research), or which I can use to test my understanding of existing mathematical phenomena (when I'm solving an exercise). Diophantine equations have proven to be a great source of number theoretic phenomena, which is why people are interested in them.

The rational root theorem and unique prime factorization are both wonderful phenomena, but ultimately are not that complicated. Thus it's natural to increase the complexity even more: what if we have a Diophantine equation which both has large degree, and which has multiple variables? This is where things get much much harder!

The simplest source of such equations is to allow yourself to have two variables x and y, and allow your Diophantine equation to be as complicated as you want in x, but force it to be *linear* in y. For example,

$$x^2 + 1 = 5y. \tag{2.1.4}$$

Note, though, that finding integer solutions to (2.1.4) is equivalent to solving

$$x^2 + 1 \equiv 0 \pmod{5}.$$

Using this trick, Diophantine equations which are *polynomial* in x and *linear* in y are always equivalent to solving polynomial equations in modular arithmetic.

So, a natural 'next step' in solving Diophantine equations is to ask how to solve equations of the form

$$p(x) \equiv 0 \pmod{N},$$

where $p(x)$ is some polynomial and N is some integer.

We are going to illustrate, through (2.1.1), how you can go about solving equations of this type.

2.2 A first reduction

Let's go back to our starting example. The Chinese remainder theorem tells us that solving

$$x^3 - 17x^2 + 12x + 16 \equiv 0 \pmod{3000}$$

is equivalent to solving the *three* different congruences

$$x^3 - 17x^2 + 12x + 16 \equiv 0 \pmod{2^3},$$

$$x^3 - 17x^2 + 12x + 16 \equiv 0 \pmod{3},$$

and

$$x^3 - 17x^2 + 12x + 16 \equiv 0 \pmod{5^3}.$$

At first it seems we've made our problem *harder*, not easier: we started with one equation, and now we have three! However, there is a sense in which our life has been made simpler: to solve

$$x^3 - 17x^2 + 12x + 16 \equiv 0 \pmod{2^3},$$

we need only plug in $x = 0, 1, 2, 3, 4, 5, 6,$ and 7. This is only eight values to try! If we simpliyfy our polynomial to

$$x^3 - 17x^2 + 12x + 16 \equiv x^3 - x^2 + 4x \pmod{8},$$

and plug in each of these values, then it's not that hard to check that the only solutions are

$$x \equiv 0, 4 \pmod{8}.$$

Similarly, we can check that the only solution to

$$x^3 - 17x^2 + 12x + 16 \equiv 0 \pmod{3}$$

is $x \equiv 1 \pmod{3}$.

Just by plugging in a 11 values, we managed to solve 2 of our 3 equations. Unfortunately, plugging in every integer between 0 and 124 to solve our equation modulo 125 is a bit too hard for me. Fortunately, Hensel found a cleverer way to handle this.

2.3 5, 25, 125

If

$$x^3 - 17x^2 + 12x + 16 \equiv 0 \pmod{5^3},$$

then $x^3 - 17x^2 + 12x + 16$ is divisible by 125. In particular, it is also divisible by 5, meaning

$$x^3 - 17x^2 + 12x + 16 \equiv 0 \pmod{5}. \tag{2.3.1}$$

(2.3.1) is much easier to solve, as we need only plug in five possibilities. If you do this, you'll find that the only solution is $x \equiv 2 \pmod{5}$.

Observe that

$$5^3 - 17 \cdot 5^2 + 12 \cdot 5 + 16 = -20$$

is divisible by 5, but not by 125.

Hensel thought that being divisible by 5 made -20 *close* to being divisible by 125. After all, -20 is divisible by five once, and once is almost three times!

On its own, this thought would probably not be so helpful. What made this so remarkable was what Hensel did *next*. Before we can explain that, though, we take a brief digression to recall an insight from calculus.

2.4 Derivatives

In calculus, one typically defines the derivative as the limit

$$f'(x) = \lim_{\epsilon \to 0} \frac{f(x+\epsilon) - f(x)}{\epsilon}.$$

It is often more intuitive to think that $f'(x)$ is the unique real number having the property that

$$f(x+\epsilon) = f(x) + f'(x)\epsilon + o(\epsilon). \tag{2.4.1}$$

Here, $o(\epsilon)$ is *little-oh* notation, and we use it to denote a quantity which is significiantly smaller than ϵ, in the sense that $o(\epsilon)$ is an error term obeying $\lim_{\epsilon\to 0} o(\epsilon)/\epsilon = 0$. But you should see $o(\epsilon)$ and think "significantly smaller than ϵ."

The point of (2.4.1) is that a derivative is an *approximation* of a function.

Analysts devised limits and little-oh notation as a way of talking about approximations and very small numbers. The idea of approximation iso fundamental that it occurs even in algebra and algebraic geometry. However, algebraic geometers have a different way of formalizing this idea. This formalization is inferior at solving problems analysts care about, but it is very useful when solving problems algebraists care about.

2.4.1 The algebraic ϵ

Consider the ring $\mathbb{R}[\epsilon]/(\epsilon^2)$, which I will often denote by just $\mathbb{R}[\epsilon]$.

This means we take the real numbers $\mathbb{R}$, and freely adjoin a new number ϵ obeying $\epsilon^2 = 0$. That is, A is the ring whose elements are of the form $x + x' \cdot \epsilon$, where $x, x' \in \mathbb{R}$, and with addition and multiplication defined as

$$(x + x'\epsilon) + (y + y'\epsilon) := (x+y) + (x'+y')\epsilon,$$

$$(x + x'\epsilon)(y + y'\epsilon) := xy + (xy' + x'y)\epsilon.$$

We should think that A is a ring containing all the usual real numbers, together with this new very small number ϵ. Consider the cubic polynomial

$$f(x) = x^3 - 17x^2 + 12x + 16.$$

In a calculus class, we'd say it's derivative is

$$f'(x) = 3x^2 - 34x + 12.$$

Using our ring $\mathbb{R}[\epsilon]$, we can discover this derivative in a different way. In the ring $\mathbb{R}[\epsilon]$, observe that

$$\begin{aligned} f(x+\epsilon) &= (x+\epsilon)^3 - 17(x+\epsilon)^2 + 12(x+\epsilon) + 16 \\ &= (x^3 + 3x^2\epsilon) - 17(x^2 + 2x\epsilon) + 12(x+\epsilon) + 16 \\ &= f(x) + \epsilon \cdot (3x^2 - 34x + 12). \end{aligned}$$

In other words, in the ring $\mathbb{R}[\epsilon]$, we literally have the equality

$$f(x+\epsilon) = f(x) + \epsilon \cdot f'(x).$$

In fact, this works for *any* polynomial function! This ring $\mathbb{R}[\epsilon]$ is perfectly suited for talking about derivatives of polynomials.

2.5 Newton's method

We now remind the reader of one more topic in standard calculus: *Newton's method*. Fix $f : \mathbb{R} \to \mathbb{R}$ some differentiable function, and suppose we want to solve $f(x) = 0$. Newton's method starts with an *approximate* solution x_0 (so $f(x_0) \approx 0$), and then produces a slightly better approximation to a solution. If everything goes well, iterating Newton's method will converge to an exact solution.

Newton's method is based on a simple observation. We know that

$$f(x_0 + \epsilon) \approx f(x_0) + f'(x_0)\epsilon.$$

So, just solve for the value of ϵ making

$$f(x_0) + f'(x_0)\epsilon = 0.$$

Then, replace x_0 by the new approximate solution $x_0 + f'(x_0)\epsilon$.

Newton's method isn't perfect (as any numerical analyst will tell you), it is a pretty decent root finding method, and works well enough for our application.

2.6 Hensel's lemma

Now, on to Hensel's lemma. Recall our problem: we wanted to solve

$$x^3 - 17x^2 + 12x + 16 \equiv 0 \pmod{125}.$$

We knew that $x \equiv 2 \pmod 5$ solves

$$x^3 - 17x^2 + 12x + 16 \equiv 0 \pmod 5.$$

This is close to what we want, but not quite.

Following Hensel, we will try to upgrade our solution mod 5 to a solution modulo 25 (and then upgrade that solution to one which exists mod 125).

Hensel made a remarkable observation. From the point of view of modern algebraic geometry, Hensel realized the following: the relationship between $\mathbb{Z}/25$ and $\mathbb{Z}/5$ is analogous to the relationship between $\mathbb{R}[\epsilon]$ and $\mathbb{R}$. More precisely,

1. the ring $\mathbb{R}[\epsilon]$ contains the number ϵ which has $\epsilon^2 = 0$, and the ring $\mathbb{Z}/25$ contains the number 5 which has $5^2 = 0$;

2. their is a surjection ring homomorphism $\mathbb{R}[\epsilon] \to \mathbb{R}$ sending ϵ to 0, and there is a surjective ring homomorphism $\mathbb{Z}/25 \to \mathbb{Z}/5$ sending 5 to 0.

We now use this observation to run Newton's method in $\mathbb{Z}/25$. Set

$$p(x) := x^3 - 17x^2 + 12x + 16.$$

Observe that $p(2) = -20$. This is our 'approximate' solution to $p(x) \equiv 0 \pmod{25}$. In what sense is it an approximate solution? Well, just as ϵ's smallness was witnessed by the fact that $\epsilon^2 = 0$, in $\mathbb{Z}/25$ we have $(-20)^2 \equiv 0 \pmod{25}$.

Now, to do Newton's method, we should replace $p(2+\epsilon)$ with $p(2+5n)$, and attempt to solve for n, using multiples of 5 as our small numbers. Observe that

$$p(2+5n) \equiv p(2) + p'(2) \cdot 5n \pmod{25},$$

by the very same manipulations we used to prove $p(x+\epsilon) = p(x) + \epsilon p'(x)$ in $\mathbb{R}[\epsilon]$.

As

$$p'(x) = 3x^2 - 34x + 12,$$

we see that $p'(2) = -44$, so that

$$p(2+5n) \equiv -20 - 220n \equiv 5 + 5n \pmod{25}.$$

To solve $5 + 5n \equiv 0 \pmod{25}$, observe that it is equivalent to solve $1 + n \equiv 0 \pmod{5}$, which is pretty easy: $n \equiv 4 \pmod{5}$. Thus the solution to $p(x) \equiv 0 \pmod{25}$ is

$$x \equiv 2 + 5 \cdot 4 \equiv 22 \pmod{25}.$$

We're now close to solving our problem – all that remains is to upgrade our solution modulo 25 into a solution modulo 125. To do this, one can use Newton's method again:

$$p(22+25n) \equiv p(22) + p'(22) \cdot 25n \pmod{125}.$$

Observe that

$$p(22) \equiv 75 \pmod{125},$$

$$p'(22) \equiv 1 \pmod{5},$$

so that

$$p(22+25n) \equiv 75 + 25n \pmod{125},$$

and thus

$$x \equiv 22 + 25 \cdot (-3) \equiv 72 \pmod{125}$$

is a solution modulo 125. And so, using ideas from calculus, Hensel was able to solve a problem in number theory!

We can state this result more abstractly as follows, for those of you into that sort of thing.

Theorem 2.6.1 (Hensel's lemma)**.** *Let $f(x) \in \mathbb{Z}[x]$ be a polynomial, and p a prime number. Suppose $a \in \mathbb{Z}$ is an integer such that $f(a) \equiv 0 \pmod p$, and such that $f'(a) \not\equiv 0 \pmod p$.*

Then, for every integer $n \geq 1$, there is a unique *congruence class $b \in \mathbb{Z}/p^n$ such that*

1. *$b \equiv a \pmod p$,*
2. *$f(b) \equiv 0 \pmod{p^n}$.*

Remark 2.6.2. It's a good exercise to try generalizing this naive form of Hensel's lemma to accomodate situations where $f(a) \not\equiv 0 \pmod p$. You can ask me if you have questions, but try thinking about it on your own for a bit first!

Proof. We prove this by induction on n, with the base case of $n = 1$ being trivial. For the induction step, assume $b \in \mathbb{Z}/p^n$ is as desired, and we want to prove there exists a unique residue $b' \in \mathbb{Z}/p^{n+1}$ with the desired conditions. Choose some integer $b \in \mathbb{Z}$ whose image in $\mathbb{Z}/p^n$ is the residue class b; by abuse of notation, we do not distinguish between the integer and the residue class.

Observe that, for any such residue b', its image in $\mathbb{Z}/p^n$ obeys the two desired properties, and so must equal b. Thus any residue $b' \in \mathbb{Z}/p^{n+1}$ obeying our constraints must be of the form

$$b' = b + p^n \cdot k$$

for some integer k. Every residue of this form automatically obeys the $b' \equiv a \pmod p$ condition. And such a residue obeys $f(b') \equiv 0 \pmod{p^{n+1}}$ if and only if

$$f(b) + f'(b) \cdot p^n \cdot k \equiv 0 \pmod{p^{n+1}}.$$

As $f(b) \equiv 0 \pmod{p^n}$, we may write $f(b) = p^n \cdot j$ for some integer j.

Observe that $f'(b) \equiv f'(a) \pmod p$ (as $b \equiv a \pmod p$), and so $f'(b)p^n \equiv f'(a)p^n \pmod{p^{n+1}}$. Thus $b' = b + p^n \cdot k$ is as desired if and only if k obeys

$$p^n \cdot (j + f'(a)k) \equiv 0 \pmod{p^{n+1}},$$

or equivalently if

$$j + f'(a)k \equiv 0 \pmod p.$$

As $f'(a) \not\equiv 0 \pmod p$, there is a unique residue $k \in \mathbb{Z}/p$ solving this equation, and hence a unique $b' \in \mathbb{Z}/p^{n+1}$ with the desired properties. □

2.7 Tangent vectors

In a calculus course, the derivative is also used to find tangents. We now explain how to think about tangent vectors in the language of algebraic geometry. For this, it's useful to focus on a concrete geometric example.

Consider the circle $x^2 + y^2 = 1$, which we will view as the prespace

$$C : \text{Ring} \to \text{Set},$$

$$C(\mathbb{R}[\epsilon]) := \{(x, y) \in A^2 \mid x^2 + y^2 = 1\}.$$

Remark 2.7.1. This prespace C is an affine scheme; in fact, $C = \operatorname{Spec} \mathbb{Z}[x, y]/(x^2 + y^2 = 1)$.

The set $C(A)$ contains all the points of $C(\mathbb{R})$. For example, $(\sqrt{2}/2, \sqrt{2}/2) \in C(A)$, and $(1, 0) \in C(A)$. But $C(A)$ also contains some new interesting points: note that $(1, \epsilon) \in C(A)$, because

$$1^2 + \epsilon^2 = 1 + 0 = 1.$$

What is the set $C(A)$? In other words, what are the quadruples of real numbers $(x, x', y, y') \in \mathbb{R}^4$ such that

$$(x + x'\epsilon)^2 + (y + y'\epsilon)^2 = 1? \tag{2.7.2}$$

The left hand side of (2.7.2) expands out to

$$(x^2 + 2xx'\epsilon) + (y^2 + 2yy'\epsilon) = (x^2 + y^2) + (2xx' + 2yy')\epsilon.$$

Thus, $C(A)$ can be described as

$$C(A) = \{(x + x'\epsilon, y + y'\epsilon) \mid x^2 + y^2 = 1, 2xx' + 2yy' = 0\}. \tag{2.7.3}$$

Remark 2.7.4. The tangent to the circle at the point (x_0, y_0) is the line

$$xx_0 + yy_0 = 1,$$

as you can compute from basic calculus. Does this remind you of (2.7.3)?

While one can profitably think about tangent vectors using tangent lines and tangent planes, there is another way to conceptualize them in algebraic geometry. The set $C(\mathbb{R}[\epsilon])$ contains the point

$$P' = \left(\frac{1}{2} - \sqrt{3} \cdot \epsilon, \frac{\sqrt{3}}{2} + \epsilon\right).$$

Intuitively, we think that this point is a small deformation of $P = (1/2, \sqrt{3}/2) \in C(\mathbb{R})$. The displacement from P to P' is thought of as an infinitesimally short vector; that infinitesimal displacement is a tangent vector, to an algebraic geometer. In other words, a tangent vector is an *infinitesimally small step which keeps you on the shape.*

2.7.1 The tangent space

Inspired by the example of $\mathbb{R}[\epsilon]$, we make the following definition.

Definition 2.7.5. Let $X : \mathrm{Ring} \to \mathrm{Set}$ be a prespace. The *tangent space* of X is the prespace

$$TX : \mathrm{Ring} \to \mathrm{Set}$$

defined by

$$TX(A) := X(A[\epsilon]/(\epsilon^2)).$$

Remark 2.7.6. For any ring A, there are natural maps

$$A \hookrightarrow A[\epsilon] \xrightarrow{\epsilon=0} A.$$

The first map is injective, and the second map is surjective (it sends $a_0 + a_1\epsilon$ to a_0).

These maps give rise to functions

$$X(A) \to X(A[\epsilon]) \to X(A),$$

which in turn give us natural transformations

$$X \to TX \to X.$$

What are these two maps? The second one, $TX \to X$, sends a tangent vector to its basepoint. The first map, $X \to TX$, is often called the *zero section*: it sends a point to the "0 vector" at that point.

Lecture 3

Lecture 4

Lecture 5

Lecture 6

Lecture 7